

Shrey Shingala

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EDUCATION

University of Waterloo

Bachelor of Software Engineering (**GPA: 4.0/4.0**)

Waterloo, ON

Sep. 2025 – Present

SKILLS

Languages: Python, TypeScript, JavaScript, SQL, C++, C, C#, Swift, Bash

Technologies: Django, React, Next.js, Node.js, PostgreSQL, Neo4j, Docker, Sentry

EXPERIENCE

Software Engineer Intern | Python, Django, Celery, TypeScript, React

May 2026 – Aug. 2026

Leap Tools

Toronto, ON

- Architected a marketing-automation engine compiling **OR-of-ANDs** trigger rules into **Django Q** objects across **4 backend services**, guaranteeing **exactly-once** delivery under concurrent Celery workers via a Postgres unique-constraint claim
- Migrated **10,000+** live loan records and their notifications to a new service behind a **per-vendor feature flag**, writing an **idempotent** cross-service backfill so both systems ran side-by-side with **zero dropped state**
- Cut the **leads** endpoint's SQL query count **83%** (124 to 21) and p95 latency **34%** by eliminating cache-bypassing calls, profiled end-to-end with **Django Silk** and **Sentry**
- Owned production observability end-to-end by building the team's **Sentry** practice from scratch, **6 dashboards**, **3** response-time **monitors**, and onboarding docs that replaced an unusable maintainer alert-email system

Software Developer Intern | Python, FastAPI, LLMs, Pandas, TypeScript

Jan. 2026 – Apr. 2026

PivoHub

(Remote) Montréal, QC

- Cut customer onboarding from **multiple days to <1 hour** by shipping an **LLM-powered** catalog import pipeline as a production **FastAPI** service, processing supplier catalogs of **1,000+ products**
- Eliminated manual data entry with an **LLM-driven** mapping engine inferring headers and emitting **structured, human-reviewable** transformation plans from messy spreadsheets at **90%+** accuracy
- Cut per-row LLM inference costs by handling **20+** extraction fields (weight, volume, pack count) in a deterministic regex layer at **95%+** accuracy, reserving embedding similarity for bilingual **29-category** classification

Software Engineer | TypeScript, Python, Pytest, Pydantic

Sep. 2025 – Present

Waterloo Rocketry

Waterloo, ON

- **Cut Electron app size 50%+** by porting a custom Python parsing library to **TypeScript** and refactoring **4 legacy parsers** from procedural code into a unified polymorphic design
- Established test infrastructure from scratch with **Pytest**, reaching **98% coverage** on bitwise decoding and CRC modules, and enforced **Pydantic** runtime validation across **60+** sensor fields to reject malformed data
- Implemented high-frequency web telemetry with a bidirectional **WebSocket-ZeroMQ bridge** in **Flask-SocketIO**, sustaining **100+ msg/s** to expose low-level rocket bus data to modern web clients

PROJECTS

Big Brother | Python, FastAPI, MongoDB, Neo4j, LLM Orchestration

[GitHub](#)

- Won **Runner-Up** at UofT Hacks 2026 by building an autonomous browser agent that **orchestrates 5 LLMs** to execute complex web tasks from natural-language intent, grounded in a **Neo4j knowledge graph** with RAG retrieval
- Reduced inference cost **40%** and average latency from **3.2s to 1.8s (44%)** with **intelligent model routing** and embedding-based semantic caching across the agent fleet

Pokémon Card Scanner | Python, FastAPI, Next.js, YOLOv8, Docker

[GitHub](#)

- Attained **95%+** identification accuracy across **20,000+** Pokémon cards by building a vision pipeline combining **YOLOv8** detection, **CLIP+FAISS** embedding search, and **OCR-based** validation
- Deployed and operated the full stack in production, **dockerized FastAPI** backend, **Next.js** frontend, Google OAuth, and **PostgreSQL**, cutting multi-card scan latency **50%** with parallel processing

EverCare | Swift, FastAPI, MongoDB, RAG

[GitHub](#)

- Built an iOS dementia-care app on a **FastAPI + MongoDB RAG** pipeline, fusing conversation history with **live biometrics** (heartbeat, breathing) for real-time distress detection
- Achieved **<2s** end-to-end voice latency with a custom audio pipeline in **Swift's AVAudioEngine**, enabling natural real-time conversations with patients